2 Localization and Dedekind domains

After a brief review of some commutative algebra background on localizations, in this lecture we begin our study of Dedekind domains, which are commutative rings that play a key role in algebraic number theory and arithmetic geometry (named after Richard Dedekind).

2.1 Localization of rings

Let A be a commutative ring (unital, as always), and let S be a multiplicative subset of A; this means S is closed under finite products (including the empty product, so $1 \in S$), and S does not contain zero. The *localization* of A with respect to S is a ring $S^{-1}A$ equipped with a ring homomorphism $\iota\colon A \to S^{-1}A$ that maps S into $(S^{-1}A)^\times$ and satisfies the following universal property: if $\varphi\colon A \to B$ is a ring homomorphism with $\varphi(S) \subseteq B^\times$ then there is a unique ring homomorphism $S^{-1}A \to B$ that makes the following diagram commute:

$$\begin{tikzcd} A \arrow[rr, "\varphi"] \arrow[dr, "\iota"'] & & B \\ & S^{-1}A \arrow[ur, dotted, "\exists!"'] & \end{tikzcd}$$

and one says that φ factors uniquely through $S^{-1}A$ (via ι). As usual with universal properties, this guarantees that $S^{-1}A$ is unique (hence well-defined), provided that it exists. To prove existence we construct $S^{-1}A$ as the quotient of $A \times S$ modulo the equivalence relation

$$(a, s) \sim (b, t) \Leftrightarrow \exists u \in S \text{ such that } u(ta - sb) = 0. \tag{1}$$

We then use a/s to denote the equivalence class of (a, s) and define $\iota(a) \coloneqq a/1$; one can easily verify that $S^{-1}A$ is a ring with additive identity $0/1$ and multiplicative identity $1/1$, and that $\iota : A \to S^{-1}A$ is a ring homomorphism. If s is invertible in A we can view a/s either as the element as^{-1} of A or the equivalence class of (a, s) in $S^{-1}A$; we have $(a, s) \sim (as^{-1}, 1)$, since $(a \cdot 1 - as^{-1} \cdot s) \cdot 1 = 0$, so this notation should not cause any confusion. For $s \in S$ we have $\iota(s)^{-1} = 1/s$, since $(s/1)(1/s) = s/s = 1/1 = 1$, thus $\iota(S) \subseteq (S^{-1}A)^\times$.

If $\varphi\colon A \to B$ is a ring homomorphism with $\varphi(S) \subseteq B^\times$, then $\varphi = \pi \circ \iota$, where π is defined by $\pi(a/s) \coloneqq \varphi(a)\varphi(s)^{-1}$. If $\pi\colon S^{-1}A \to B$ is any ring homomorphism that satisfies $\varphi = \pi \circ \iota$, then $\varphi(a)\varphi(s)^{-1} = \pi(\iota(a))\pi(\iota(s))^{-1} = \pi(\iota(a)\iota(s)^{-1}) = \pi((a/1)(1/s)) = \pi(a/s)$, so π is unique.

In the case of interest to us, A is actually an integral domain, in which case $(a, s) \sim (b, t)$ if and only if $at - bs = 0$ (we can always take $u = 1$ in the equivalence relation (1) above), and we can then identify $S^{-1}A$ with a subring of the fraction field of A (which we note is the localization of A with respect to $S = A_{\neq 0}$), and if T is a multiplicative subset of A that contains S, then $S^{-1}A \subseteq T^{-1}A$.

When A is an integral domain the map $\iota\colon A \to S^{-1}A$ is injective, allowing us to identify A with its image $\iota(A) \subseteq S^{-1}A$ (in general, ι is injective if and only if S contains no zero divisors). When A is an integral domain we may thus view $S^{-1}A$ as an intermediate ring that lies between A and its fraction field: $A \subseteq S^{-1}A \subseteq \operatorname{Frac} A$.

2.2 Ideals in localizations of rings

If $\varphi\colon A \to B$ is a ring homomorphism and $\mathfrak{b}$ is a B-ideal, then $\varphi^{-1}(\mathfrak{b})$ is an A-ideal called the *contraction* of $\mathfrak{b}$ to A and sometimes denoted $\mathfrak{b}^{\mathrm{c}}$; when A is a subring of B and φ is